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ORIGINAL ARTICLE

A robotic polishing parameter optimization method considering time-varying wear

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Abstract

In the automatic polishing process, the wear of polishing tools and the change of polishing parameters will affect the Preston coefficient, which makes it difficult to establish an accurate material removal model to achieve stable and excellent polishing quality. In this paper, a robotic polishing parameter optimization method considering time-varying wear is proposed to address these issues. First, combining the rich information in the theoretical modeling method with the data-driven regression method, a material removal regression model incorporating prior knowledge is proposed, which greatly reduces the large amount of experimental data required by the original regression model. The proposed model is able to track the wear variation of the sandpaper as well as the effect of polishing parameters. Then, based on the proposed prediction model, the genetic algorithm is used to optimize the polishing parameters in order to achieve better machining quality and less energy consumption. Finally, the experimental verification is carried out on the hybrid robot polishing test bench. The results show that the proposed material removal regression model incorporating prior knowledge has higher prediction accuracy and less required experimental data than existing models. The proposed robot polishing parameter optimization method can effectively compensate for tool wear and ensure the consistency of material removal during polishing while reducing energy consumption.

Keywords Time-varying wear · Gaussian process regression · Material removal model · Genetic algorithm · Parameter optimization

1 Introduction

Mold polishing is a very important but time-consuming and costly work in the mold manufacturing process [1]. It can remove the convex part of the surface caused by machining and improve the surface quality of the mold. Nowadays, most of the mold polishing mainly relies on manual processing by skilled workers, which is a kind of low-efficiency and repetitive labor with large randomness. Manual polishing is difficult to ensure the stability and consistency of processing quality. In addition, these processes expose laborers to an environment of high noise and metal dust that is harmful to their health [2]. Therefore, it is of great significance to carry out research on polishing automation.

In the automatic polishing process, robots can use a variety of polishing tools to polish different types of workpieces and achieve uniform removal by adjusting parameters such as polishing force, rotation speed, and feed rate. However, how to select appropriate polishing parameters to achieve the minimum energy consumption under the premise of ensuring polishing quality is always a difficult point, which requires an accurate material removal model as the basis for parameter selection. Unfortunately, due to the inevitable wear of polishing tools during processing and the complexity of the coupling interplay of mechanical, the construction of an accurate material removal model is a great challenge.

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At present, there are mainly two types of material removal model modeling methods, one is the theoretical modeling method, and the other is the regression modeling method.

The theoretical modeling method refers to establishing models based on ideal physical contact theories such as the Preston equation [3] and Hertzian contact theory [4]. For the wear problem of polishing tools, Borucki [5] proposed a mathematical model of the gradual decay of material removal with wear using the Greenwood-Williamson theory of contact, which attributed the material removal decay to the change of pad roughness and built the foundation of many modeling approaches. Lee et al. [6] analyzed the effect of wear on the microstructure of the polishing disc but only made a qualitative analysis. Furthermore, Cao et al. [7] conducted a multi-scale study on the polishing mechanism, then studied the influence of the surface morphology (roughness distribution) of the polishing tool on the material removal rate from the microscopic scale, and the compensation coefficient obtained from this can well compensate for the tool wear. The variation of polishing contact area during polishing pad wear was taken into account in the model by Shi et al. [8], which improved the prediction accuracy of the model. Wan et al. [9] transformed the wear rate of the polishing pad into the wear rate of a single abrasive grain multiplied by the number of effective active abrasive grains per unit area from a microscopic perspective, it revealed how the pad surface changes with the pad wear and quantitatively calculated material removal function. However, in the actual processing, the change of wear is dynamic and time-varying, so it is difficult to obtain some information such as the surface topography of the polishing tool in real-time to evaluate the degree of wear. Besides, these aforementioned theoretical models all regard the Preston coefficient k_p as an independent constant with the velocity distribution and pressure distribution, which is inconsistent with the actual experiments [10, 11]. Therefore, although the theoretical model can accurately predict the material removal rate in a short period by comprehensively considering the influencing factors and certain experimental data, the prediction accuracy of the theoretical model will gradually decrease with the change of time and various variables.

The regression modeling method refers to fitting the material removal rate using mathematical statistics and machine learning methods. Yang et al. [12, 13] proposed a grinding process modeling method based on the support vector machine (SVM). This regression model uses feed rate, contact force, and surface curvature parameters to predict grinding removal, and its prediction accuracy is approximately 90%. Similarly, Yue et al. [14] applied SVM to grinding depth prediction. Although the model can modify the model according to the measurement data, its experimental cost and time are excessive due to the large amount of required data. Pandiyan et al. [15] evaluated different statistical regression methods using the data obtained from the Taguchi experiment then concluded that the effect of sand particle size had the greatest impact on material removal, but did not consider the change in sand particle size due to wear during grinding. A deep belief network [16] and GMDH-type polynomial neural networks [17] were employed to predict material removal rate based on the acting force distribution and grinding parameters. Additionally, Mohammad et al. [18] presented an incorporation of neural networks and genetic algorithms to optimize the polishing process. Even so the article achieves the removal of uneven surfaces, it ignores the problem of tool wear. Feng et al. [11] provided a method to improve the accuracy of the classic model in bonnet polishing by applying Gaussian process regression. The article mainly focuses on the coupling relationship between the Preston coefficient and polishing depth and rotation speed. Xu et al. [19] used neural networks to study the effect of slurry composition concentration on material removal rate. Due to the limited amount of training data, the model prediction performance is relatively ordinary. In general, using regression modeling is a more effective method to quantify polishing tool wear changes and the coupling relationship between Preston coefficients and polishing parameters than theoretical modeling methods. It can capture the main variables and use experimental data to get direct results. However, the "black box" algorithm used in the existing regression modeling basically does not use the physics and geometry knowledge of the existing theoretical model, which leads to a large input dimension (including rotation speed, feed rate, polishing pressure, polishing angle, tool radius, surface curvature, and processing time). The amount of data required for regression increases exponentially to obtain a great removal model. Consequently, the time and cost of modeling is huge, which obviously does not match our actual manufacturing needs. Moreover, few studies have considered both the Preston coefficient and the tool wear.

The objective of this study is to provide a polishing parameter optimization method that can compensate for the wear of polishing tools, so that the material removal rate is more consistent in the whole process and the polishing quality is better. First, in view of the advantages and disadvantages of the above two modeling methods, a material removal regression model combined with prior knowledge is proposed. By taking the relationship between material removal depth and polishing parameters in the theoretical model as prior knowledge and using the Gaussian process regression method to quantify the Preston coefficient k_p, the proposed method enables accurate prediction of material removal with less experimental data. Then, based on the proposed model, the polishing parameters can be optimized by using the genetic algorithm. It aims to achieve consistent material removal while minimizing energy consumption.

The remainder of this study is organized as follows. "Sect. 2" presents the material removal regression model. The optimization method is proposed in "Sect. 3." The experimental validation and analysis of the proposed method are detailed in "Sect. 4." Finally, the paper is summarized in "Sect. 5."

2 Material removal regression model incorporating prior knowledge

The proposed model combines the advantages of the theoretical model and the regression model. It transforms the constant term Preston coefficient in the theoretical model into a time-varying function that changes with the polishing process. In the theoretical model, the internal relationship between polishing parameters can be taken as the prior knowledge of the regression model. The tool wear is able to be quantified by using Gaussian process regression, and then accurately calculate material removal depth, which lays the foundation for robotic polishing parameter optimization.

2.1 Theoretical model

The Preston equation was proposed in 1927 [3] and is extensively used in current polishing processes. It describes the material removal rate as the following formula:

$$\mathrm{d}h = k_{\mathrm{p}} P(x,y) V(x,y) \mathrm{d}t \tag{1}$$

where $\mathrm{d}h$ is the removal depth in unit time $\mathrm{d}t$. $P(x,y)$ denotes the contacting pressure on the contact area. $V(x,y)$ denotes the relative velocity on the contact area, x and y are the coordinate position of the polishing contact point. k_{p} denotes the Preston coefficient.

Since the robot-controlled polishing experiment actually adopts the method of controlling the polishing disc to continuously feed along the polishing path, use the Preston equation in the form of velocity and rewrite Eq. (1) as follows:

$$\mathrm{d}h = k_{\mathrm{p}} \frac{P(x,y)V(x,y)}{v_{\mathrm{f}}} \mathrm{d}l \tag{2}$$

where v_{f} denotes the feed rate along the path.

Since the modeling method proposed in this paper applies the $P(x,y)$ and $V(x,y)$ in the existing theoretical model and transforms the Preston coefficient k_{p} into a time-varying function with wear and polishing parameters, this method is suitable for the processing of different polishing tools such as polishing with disc tools, spherical tools, and cylindrical tools. All we need to do is to replace the $P(x,y)$ and $V(x,y)$ in the corresponding theoretical model. Without loss of generality, the paper studies polishing using disc tools.

Here, we establish the $P(x,y)$ and $V(x,y)$ according to the work of Li et al. [20]. As shown in Fig. 1, when the polishing disc moves on a plane along a straight line, the feeding direction is the y-direction, we can get the pressure at a point $M(x,y)$ in the polishing area as follows:

$$P(x,y) = \frac{F_{\mathrm{n}}}{\pi {R_{\mathrm{p}}}^2} \tag{3}$$

where F_{n} is the normal positive force of the disc, and R_{p} is the radius of the polishing disc.

The relative linear velocity distribution of $M(x,y)$ is

$$V(x,y) = \sqrt{v_{\mathrm{f}}^2 + v_{\mathrm{m}}^2 + 2v_{\mathrm{f}}v_m \cos\alpha} = \sqrt{v_{\mathrm{f}}^2 + {\omega_{\mathrm{p}}}^2(x^2+y^2) + 2v_{\mathrm{f}}\omega_{\mathrm{p}}x} \tag{4}$$

where ω_{p} denotes the rotational angular velocity of the disc, $v_{\mathrm{m}} = \omega_p\sqrt{x^2+y^2}$ in Fig. 1 denotes the linear velocity corresponding to the rotation, and α is the angle between v_{f} and v_{m} and $\cos\alpha = \frac{x}{\sqrt{x^2+y^2}}$.

Since the polishing process of the robot is an accumulation process, the material removal at a certain point M in the polishing contact area can be regarded as the accumulation from the L_2 point to the L_1 point. Accordingly, we can obtain the Preston equation of the polishing disc along the trajectory:

$$\begin{aligned} h &= \int_{L_2}^{L_1} \frac{\mathrm{d}h}{\mathrm{d}l}\mathrm{d}y \\ &= \frac{k_{\mathrm{p}}}{v_{\mathrm{f}}} \int_{y_2}^{y_1} \frac{F_{\mathrm{n}}}{\pi {R_{\mathrm{p}}}^2} \times \sqrt{v_{\mathrm{f}}^2 + \omega^2(x^2+y^2) + 2v_{\mathrm{f}}\omega x}\mathrm{d}y \end{aligned} \tag{5}$$

Through the experimental determination of the Preston coefficient k_{p}, an accurate material removal model for the new polishing tool can be obtained. However, in the actual

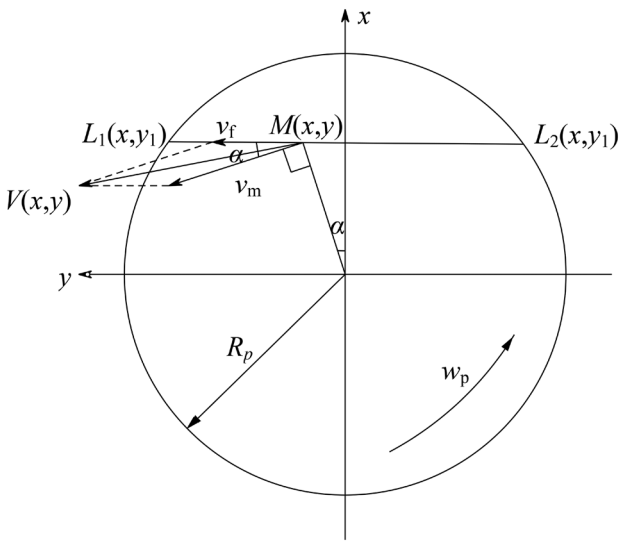


Fig. 1 The relative velocity distribution

polishing process, the polishing tool will gradually wear out until it fails, and k_p will also change accordingly, making the resulting removal model invalid. In addition, polishing parameters also affect k_p. Therefore, it is necessary to study the changing process of polishing tool wear and the influence of polishing parameters on k_p from a data-driven perspective. Note that some other regression methods can make effective predictions, this paper chooses GPR (Gaussian process regression) for the following reasons:

1. The GPR model is based on statistical theory, so the model can not only provide the predicted mean function but also give the model credibility.
2. Compared with the neural network method, since the covariance function is pre-specified to describe the model characteristics, GPR can provide better prediction results when the number of samples is small.
3. The properties of the covariance function make the model easy to extend and can describe complex features at multiple scales. In this paper, only the feed speed, polishing force, and time are considered. If we consider the rotation speed, sandpaper particle size, and other factors, all we need to do is to design an appropriate covariance function to characterize the corresponding features.

2.2 Gaussian process regression modeling

2.2.1 Gaussian process regression theory

GPR is a non-parametric probabilistic model that uses Gaussian process prior to perform regression analysis on data. It is widely used in time series analysis due to the convenience of its kernel function and the ability to provide posteriors for prediction results [21]. For given training data $D = \{(X_1, Y_1), (X_2, Y_2), ..., (X_m, Y_m)\}, X_i, Y_i \in R$, the GPR problem can be described as:

$$y(X) = f(X) + \varepsilon \tag{6}$$

where $X = [X_1, X_2, ..., X_m]$, $y(X)$ denotes the measured data, ε is the correlated error of the measured data, and $f(X)$ is the unknown function to be estimated. As shown in Eqs. (7)–(9), $f(X)$ satisfies a Gaussian distribution with a mean function of $m(X)$ and a covariance function of $K(X, X')$.

$$f(X) \sim N\left(m(X), K(X, X')\right) \tag{7}$$

$$m(X) = E\left[f(X)\right] \tag{8}$$

$$K(X, X') = E\left[(f(X) - m(X))(f(X') - m(X'))\right] \tag{9}$$

The covariance function $K(X, X')$ is the key to GPR, it evaluates the correlation between predictors and determines how the response at one point X_i is affected by responses at other points X_j. We need to choose a combination of one or more kernel functions to fit different features. These kernel functions are parameterized by hyperparameter vectors, so the kernel functions are often expressed as $K(X_i, X_j|\theta)$. Common kernel functions include squared exponential function, Matern function, automatic relevance determination, and periodic function.

The hyperparameter vector θ is determined by the maximization of the marginal log-likelihood function shown in Eq. (10) based on the given training set D.

$$\ln p(Y|X, \theta) = -\frac{1}{2}\ln\left|K + \sigma_n^2 I\right| - \frac{1}{2}Y^T(K + \sigma_n^2 I)^{-1}Y - \frac{n}{2}\ln(2) \tag{10}$$

where K is the covariance function $K(X, X')$, and σ_n^2 is the covariance of the observation noise. Using the calculated covariance function K, we can obtain the joint probability density distribution of the data to be predicted and the original training dataset as shown below:

$$\begin{bmatrix} Y \\ f^* \end{bmatrix} \sim N\left(\begin{bmatrix} m(X) \\ m(X^*) \end{bmatrix}, \begin{bmatrix} K(X,X) + \sigma_n^2 I & K(X,X^*) \\ K(X^*,X) & K(X^*,X^*) \end{bmatrix}\right) \tag{11}$$

Finally, the mean m^* and variance $\mathrm{cov}(f^*)$ are obtained by calculating the marginal distribution of f^* as follows:

$$f^* \mid X^*, X, Y \sim N(m(X^*), \mathrm{cov}(f^*)) \tag{12}$$

$$m^* = m(X^*) + K(X^*, X)(K(X, X) + \sigma_n^2 I)^{-1}(Y - m(X)) \tag{13}$$

$$\mathrm{cov}(f^*) = K(X^*, X^*) - K(X^*, X)(K(X, X) + \sigma_n^2 I)^{-1}K(X, X^*) \tag{14}$$

2.2.2 Wear quantification and material removal prediction

In this paper, we only study the variation of the Preston coefficient with polishing force, feed rate, and time, so X in GPR is $\left[F_n \;\; v_f \;\; t\right]^T$, and $y(X)$ is the material removal depth h that needs to be predicted. To obtain the experimental data required for the model construction, the experiment was carried out on the hybrid robot polishing test bench, and the contour of the specimen was collected by the Nanovea 3D optical surface profiler. Due to the acquisition limitation of the profilometer, the experimental specimens were designed as flat steel plates made of 5CrNiMo with a length and width of 80 mm. As shown in Fig. 2, during the actual experiment, in order to ensure the continuity of the polishing process, three specimens were placed side by side, and the experimental video is shown in Online Resource 1.

The experimental process of establishing a GPR model incorporating prior knowledge to predict material removal is as follows:

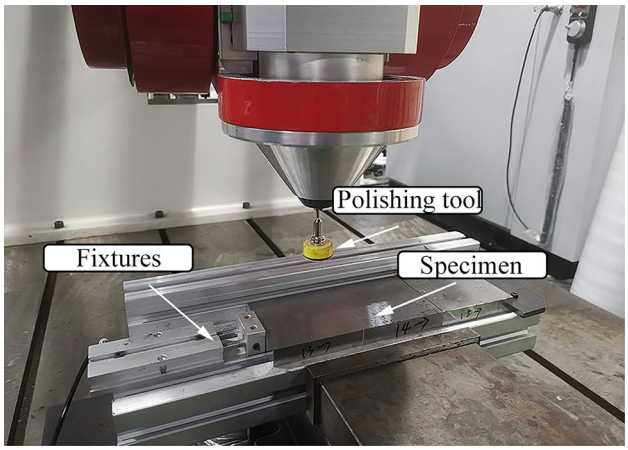


Fig. 2 The experimental setup

1. Do the trajectory planning as shown in Fig. 3 for the three specimens fixed side by side. In the figure, O_1 is the starting point of the trajectory, O_2 is the end point, and the arrows indicate the polishing feed direction.
2. Obtain the initial contour height h_0 of the workpiece surface by scanning with the 3D profiler. The measurement ranges of the three steel plates were points A to B, B to C, and C to D in Fig. 3.
3. In each experiment, the normal polishing force and feed rate were kept constant during the same polishing process. Before the formal experiment, we conducted a series of preliminary experiments to find the optimal range of polishing parameters. First, regarding the determination of the polishing force constraint range, it is found that when the polishing force F_n is too low (10 to 25 N), the polishing effect is poor although the sandpaper wears very slowly. And when the polishing force F_n is too large (55 N and above), the abrasive particles of the sandpaper are too embedded in the mold, which is easy to cause surface damage and over-polishing. As for the feed rate v_f, to ensure the processing efficiency, the feed rate v_f is not suitable for smaller than 1 mm/s. Besides, when the feed rate v_f is greater than 4 mm/s, the polishing tool cannot effectively remove the material due to the short contact time with the mold. As a result, we choose 30–50 N as the constraint range of polishing force F_n, and 1–4 mm/s as the constraint range of feed rate v_f. Table 1 shows the main experimental parameter settings.
4. Measure the contour height h' after polishing again. Figure 4 shows the surface morphology of the specimen before and after polishing measured by the second pass through the BC segment in Fig. 4 under the conditions of $F_n = 40$ N and $v_f = 2$ mm/s. The h_i and h_i' ($i = 1, 2, 3, 4, 5$) marked in the figure are the surface contour height of the workpiece before and after polishing at some sampling points. We can calculate the corresponding material removal depth h by subtracting them. Polishing time is calculated by dividing the distance by the feed rate.
5. Figure 5 shows the scatter plot of the maximum removal depth with time under the condition of $F_n = 40$ N and $v_f = 2$ mm/s. It can be seen from the figure that the maximum removal depth generally decreases due to wear. Additionally, referring to the literature [11], the coupling relationship between polishing parameters has long-term smooth features and short-term periodic features, which can be characterized by setting a composite kernel function with SE (square exponential) kernel and PER (periodic) kernel.

$$K_1(X,X') = \theta_1^2\, exp(-\frac{(X-X')^2}{2\theta_2^2}) + \theta_3^2 \exp(-\frac{2}{\theta_4^2}\sin^2(\frac{\pi(X-X')}{T})) \tag{15}$$

where θ_1, θ_2, θ_3, θ_4, and T are the hyperparameters.

Excluding abnormal points that deviate significantly from the distribution, the data presents a certain Gaussian distribution due to observation errors, so the noise model can be specified as the sum of a squared exponential contribution and an independent component as follows:

$$K_2(X,X') = \theta_5^2 \exp(-\frac{(X-X')^2}{2\theta_6^2}) + \theta_7^2 \delta_{XX'} \tag{16}$$

where θ_5 is the magnitude of the correlated noise component, θ_6 is its length scale, and θ_7 is the magnitude of the independent noise component, which are also hyperparameters. The noise of the system can be caused by observational errors as well as changes in the robotic polishing process, so it is reasonable to assume some temporal correlation in addition to the independent noise components.

The final covariance function is:

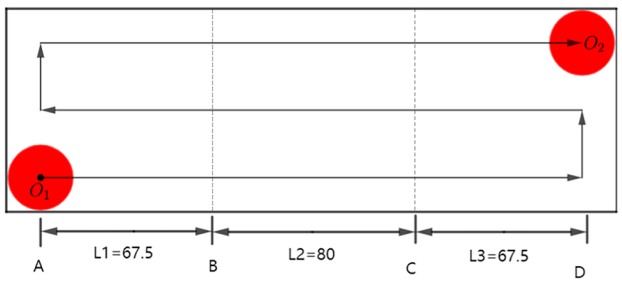


Fig. 3 Experimental trajectory planning

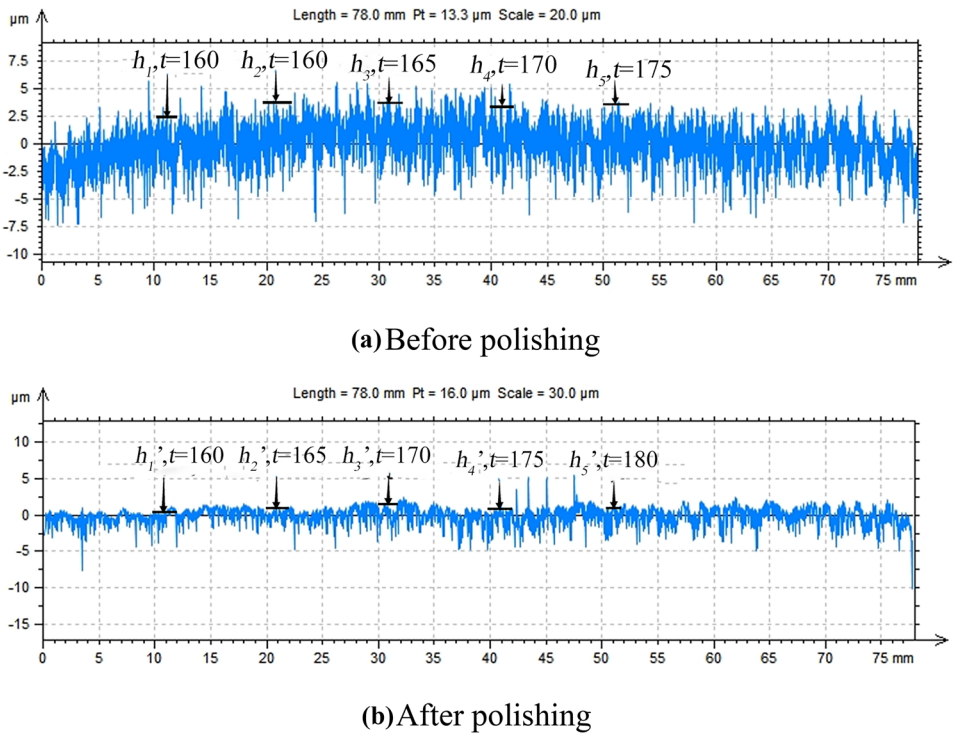


Fig. 4 The surface profile of specimen before and after polishing

$$K(X,X') = K_1(X,X') + K_2(X,X') \tag{17}$$

After setting a suitable covariance function, train the model. Figure 6 shows the algorithm flow chart. The identity transformation of Formula (5) is carried out, then we can get the following equation:

$$k_p = \frac{hv_f}{\int_{y_2}^{y_1} \frac{F_n}{\pi R_p^2} \times \sqrt{v_f^2 + \omega^2(x^2+y^2) + 2v_f\omega x}\,dy} \tag{18}$$

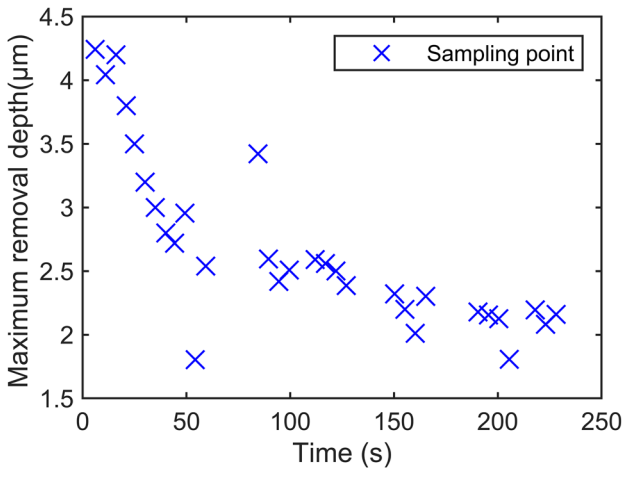


Fig. 5 Scatter plot of maximum removal depth under the condition of F_n=40 N, v_f=2 mm/s

Before training the GPR model, this equation is input into the model as prior knowledge, so that the relationship between the material removal depth h and time is transformed into the relationship between the Preston coefficient

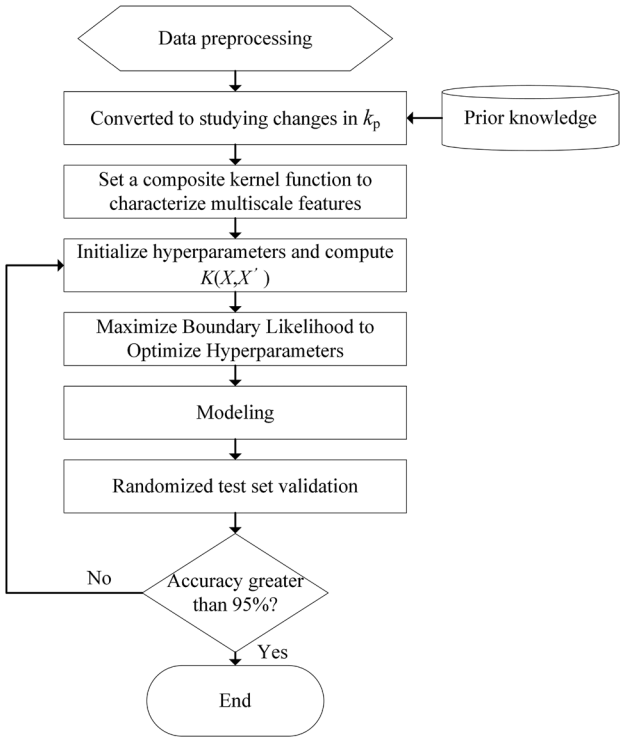
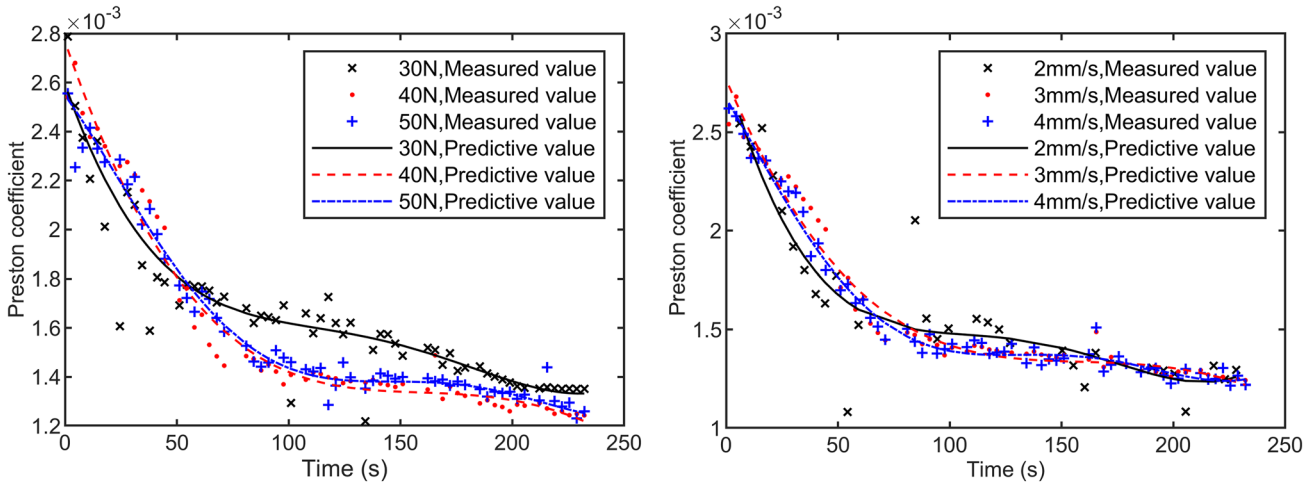


Fig. 6 Algorithm flow chart

k_p and time, which simplifies the prediction problem. To prevent the optimization algorithm from falling into a local optimal solution, a few random restarts were tried, picking the run with the best marginal likelihood. When the difference between the actual measured material removal depth and that predicted by the GPR model is within the interval of ±0.05μm, the prediction of the GPR model is considered correct. The training accuracy in Fig. 6 refers to the proportion of correct points predicted by the GPR model in the test set. Considering some problems such as stains on the specimen or vibration of the measuring equipment during the measurement process, some "abnormal points" that deviate significantly from the ideal value will be obtained in the experiment. Therefore, when the accuracy rate is greater than 95%, the trained model is considered feasible. In addition, since the polishing force and feed rate remain unchanged in each experiment, to ensure that the model is still applicable in the case of changing parameters and further learn the coupling relationship between polishing parameters, a small number of training samples with changing parameters were added.

Figure 7a shows the variation curve of k_p with polishing time under different polishing forces at the same feed rate of 3 mm/s obtained through experiments. Figure 7b shows the variation curve of k_p with polishing time under the same polishing force of 40 N and different feed rates. It can be seen from Fig. 7 that in the early stage of polishing, the sandpaper wears quickly and k_p decreases rapidly because the abrasive grains of the sandpaper are relatively new and they are prone to fall off and break. After polishing for a while, the wear enters a stable period and the wear rate slows down significantly, which is approximately linear. When the wear reaches a certain level, the polishing ability of the sandpaper almost completely fails, and it no longer has the polishing ability. More specifically, it can be seen from Fig. 7a that the polishing force not only affects the k_p at the initial moment but also affects the time when the sandpaper enters the stable period and the corresponding value of the k_p. The results in Fig. 7b show that changes in feed rate have little effect on k_p. Since the feed rate is much smaller than the speed brought by the high-speed rotation of the spindle, it has little effect on the speed distribution of the polishing model, which explains the insensitivity of k_p to the changes in feed rate. Therefore, in practical use, the influence of the feed rate can be ignored within a certain range of process constraints, which further reduces the experimental cost and required time.

By substituting the polishing parameters, we can obtain how the material removal depth changes with time under certain working conditions. Figure 8 shows the change of material removal depth under the condition of $F_n = 40$ N and $v_f = 2$ mm/s, the length means the contact length between the sandpaper and the workpiece. It can be seen from the figure that the material removal amount gradually decreases with the wear of the sandpaper. To compensate for the wear of the sandpaper and ensure the consistency of the material removal amount, polishing parameters should be optimized during the actual polishing process. Compared with the influence of the rotation speed on k_p, k_p is less sensitive to changes in force and feed rate [11]. Through experiments, it is found that the sandpaper will quickly wear to failure when the rotation speed is too high, making it difficult to work for a long time; if the rotation speed is too small, although the wear of the polishing tool is slowed down, the polishing effect is poor. As a result, in this paper, polishing force and feed rate were selected as polishing parameters to compensate for wear. In order to achieve better polishing quality,



(a) At the same feed rate of 3 mm/s and different forces

(b) At the same forces of 40 N and different feed rate

Fig. 7 Preston coefficient with time under different polishing conditions

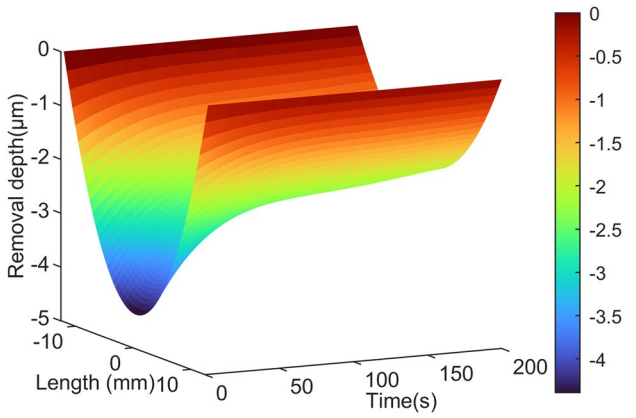


Fig. 8 Removal depth prediction under the condition of $F = 40$ N and $v_f = 2$ mm/s

it was also necessary to determine an appropriate rotation speed.

2.2.3 Model comparison analysis

The method proposed in this paper was trained using four training sets with different sample sizes to determine the minimum number of training samples required for a good prediction performance. The number of training samples for the four sets was 300, 450, 600, and 900, respectively. Forty randomly selected sample points were used as the test set to test and compare the performance of the models. Before outputting the results, sort the test set according to the measured values from small to large. The prediction results are shown in Fig. 9. The small picture in the lower right corner of Fig. 9 is a partially enlarged picture. From the results, we can see that the prediction performance of the model with 450 training samples has been relatively good, with the increase in the number of training samples,

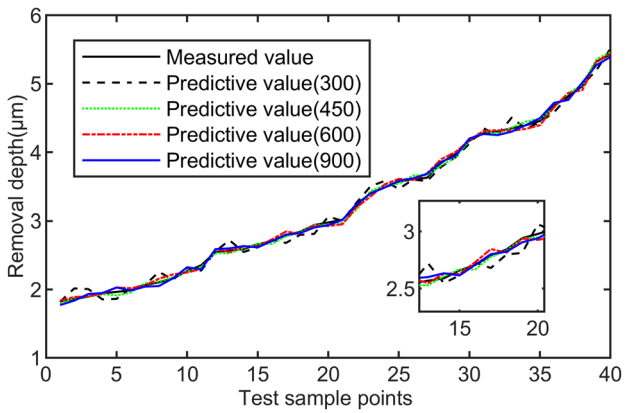


Fig. 9 Removal depth prediction results for the proposed model with different sample sizes

and the prediction performance of the proposed method is almost the same. Taking into account the reduction of experimental cost and time, a training set with 450 sample points can meet the prediction accuracy requirements, and then this model can be used to compare with other algorithms.

To verify the superiority of the material removal regression model incorporating prior knowledge proposed in this paper, the proposed model is compared with the GPR model proposed by [11], the support vector regression (SVR) model proposed by [12], and the theoretical model proposed by [20]. The test samples were also 40 data points randomly selected from other historical experimental groups. Rearrange the samples from small to large as before. Figure 10 shows the comparison results of the above algorithms. In Fig. 10, model 1 is the GPR method proposed in this paper that integrates prior knowledge, and the number of training samples is 450; model 2 is the SVR model with 900 training samples. Since the rotation speed, tool radius, and some other factors are kept constant in this experiment, the model input and output are same with this paper. The nonlinear mapping function $h = f(X) = w^T \varphi(X) + b$ is solved using the data of the training set to achieve the prediction; model 3 is the SVR model with 1200 training samples; model 4 is the GPR model proposed by [11], which only considers the influence of polishing parameters on k_p, but does not consider the change of k_p with time due to wear. The theory of GPR is described in "Sect. 2.2.1." Select the SE kernel and PER kernel as the covariance function, then use the modified k_p (constant) to calculate the removal depth; model 5 is the theoretical model. k_p (constant) was calculated by Eq. (5) under the condition of $F_n = 40$ N, $v_f = 2$ mm/s. The residual sum of squares (RSS), root mean square error (RMSE), mean absolute percentage error (MAPE), and mean absolute error (MAE) were used as error evaluation indicators. The error analysis results are shown in Table 2.

From Fig. 10 and Table 2, we can see that most of the predicted values of model 5 (theoretical model) were larger than the measured values, resulting from the theoretical model treats k_p as constant and did not account for wear. The error of this model depends on polishing time. The prediction accuracy of model 4 (GPR model proposed by [11]) has been improved compared with the theoretical model, but the degree of improvement is limited. This can be explained by the fact that the effect of wear is much larger than the effect of polishing parameters for k_p. Model 2 can also predict material removal depth to a certain extent, but the prediction on some data points has a large deviation from the measured value. Compared with model 2, the RSS, RMSE, MAPE, and MAE of model 1 are reduced by 0.0161 μm^2, 0.0497 μm, 1.46%, and 0.042 μm, respectively. The reason is that the SVR method used in model 2 is a pure black-box method. When the number of training samples is

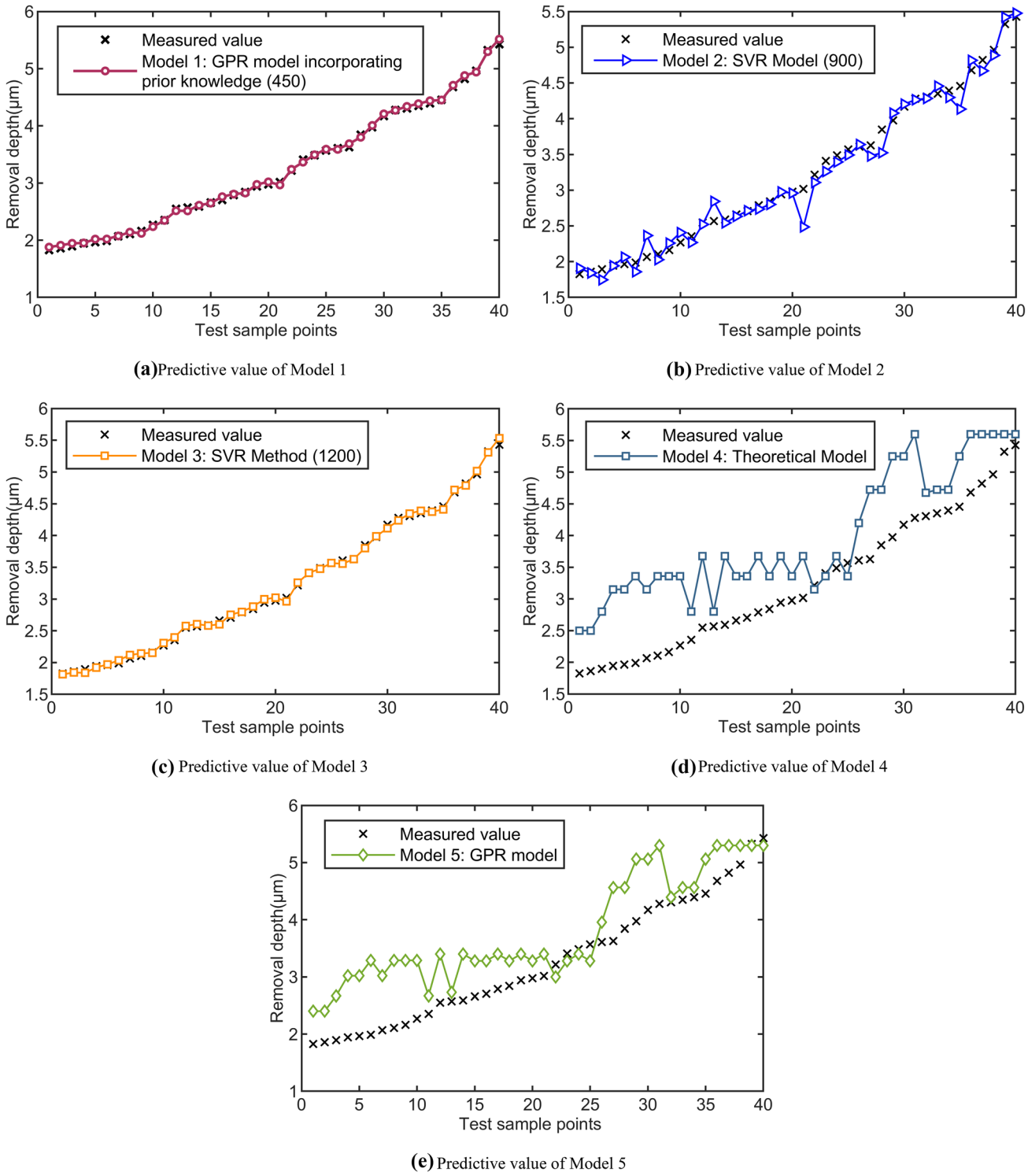


Fig. 10 Removal depth prediction results for different models

600, the samples are relatively sparse for the distribution of model inputs, which is not enough to completely fit the correct material removal. Especially for unknown samples, model 2 has poor predictive power. When the number of training samples increases to 1200, it can be found from Table 2 that the error evaluation indexes of the model 3 and model 1 are not much different. It can be considered that the prediction ability of the two models is basically

Table 1 The main experimental parameter settings

Polishing tool material	The radius of polishing disc (mm)	Rotation rate (rpm)	Sandpaper grit	Polishing force (N)	Feed rate (mm/s)
Replaceable sandpaper	25	2500	600	30–50	1–4

the same. However, the number of training samples of the method in this paper is only 450, which is much less than the SVR method.

It is worth mentioning that the speed distribution and pressure distribution contained in the theoretical model of disc plane polishing used in this paper are still relatively simple, and this paper only considers force and feed rate as variables. When we take into account more factors such as polishing inclination, surface curvature, and rotation speed, the dimension of the input matrix of the pure black-box method will greatly increase, which will lead to an exponential increase in the number of experimental samples required. Furthermore, the pure black-box approach can only be remodeled when changing polishing tools, but the method proposed in this paper does not require additional experiments as long as a suitable theoretical model is adopted and the corresponding variables are brought into the model. When polishing tools are changed, less experimental time and cost are required since only k_p needs to be studied. In conclusion, the GPR model incorporating prior knowledge proposed in this paper has the highest prediction accuracy and less required data.

3 Polishing parameters optimization

As mentioned above, for the purpose of better surface quality, it is necessary to compensate for the wear of the polishing tool by adjusting the polishing parameters (feed rate v_f and polishing force F_n). We can only regulate a certain polishing parameter such as feed rate v_f (or polishing force F_n), but considering the large amount of energy consumed by polishing, it is worthwhile to reduce the energy consumed while ensuring the polishing performance. In this section, parameter optimization methods considering the minimum energy consumption are introduced. "Sect. 3.1" describes the calculation of the energy consumed by polishing, and "Sect. 3.2" presents the polishing parameter optimization algorithm.

Table 2 Error analysis results comparison

Models	RSS/μm^2	RMSE/μm	MAPE/%	MAE/μm
Model 1	0.0600	0.0387	1.16%	0.0341
Model 2	0.9289	0.1524	3.69%	0.1095
Model 3	0.0655	0.0405	1.14%	0.0345
Model 4	27.5327	0.8296	26.83%	0.7361
Model 5	18.9395	0.6881	22.14%	0.5882

3.1 Energy consumption calculation

As shown in Eq. (19), the energy consumption can be represented by the mechanical work done by all the motors [22].

$$E = E_1 + E_2 = \sum_{i=1}^{n} \int_0^t \tau_i \omega_i \mathrm{d}t + \int_0^t M\omega_n \mathrm{d}t \tag{19}$$

where E represents the mechanical work done by all the motors of the robot, E_1 represents the mechanical work done by the motors of each joint, and E_2 represents the mechanical work done by the high-speed rotation of the end motorized spindle. τ_i is the joint torques of motor i, ω_i is the angular motor velocity of motor i, and M is the friction torque of the plane polishing disc. t refers to the time it takes to pass the specified path.

The joint torques can be obtained by the rigid body dynamics model of the robot as follows:

$$\tau = D(q)\ddot{q} + H(q,\dot{q})\dot{q} + g(q) \tag{20}$$

where $D(q)$ is a positive definite mass matrix, $H(q,\dot{q})$ is a matrix considering Coriolis, and $g(q)$ denotes the vector accounting for gravity.

As shown in Fig. 11, using the micro-element analysis method can give the magnitude of the frictional force on the surface element of the disc as shown in Eq. (21).

$$\mathrm{d}f = \mu \frac{F_n}{\pi {R_p}^2 r \mathrm{d}r \mathrm{d}\,\theta} \tag{21}$$

The velocity of the unit relative to the table surface can be expressed as:

$$\boldsymbol{v} = (v_f + r\omega_n \sin\theta)\boldsymbol{i} - r\omega_n \cos\theta\, \boldsymbol{j} \tag{22}$$

Because the direction of the friction force on the surface element is opposite to the direction of the speed, the friction force on the surface element is:

$$\mathrm{d}f = -\mu \frac{F_n}{\pi {R_p}^2} \frac{(v_f + r\omega_n \sin\theta)\boldsymbol{i} - r\omega_n \cos\theta\, \boldsymbol{j}}{\sqrt{(v_f + r\omega_n \sin\theta)^2 + r^2 {\omega_n}^2 \cos^2\theta}} \mathrm{d}r\,\mathrm{d}\,\theta \tag{23}$$

In this way, the friction torque M borne by the entire disc can be obtained by Eq. (24).

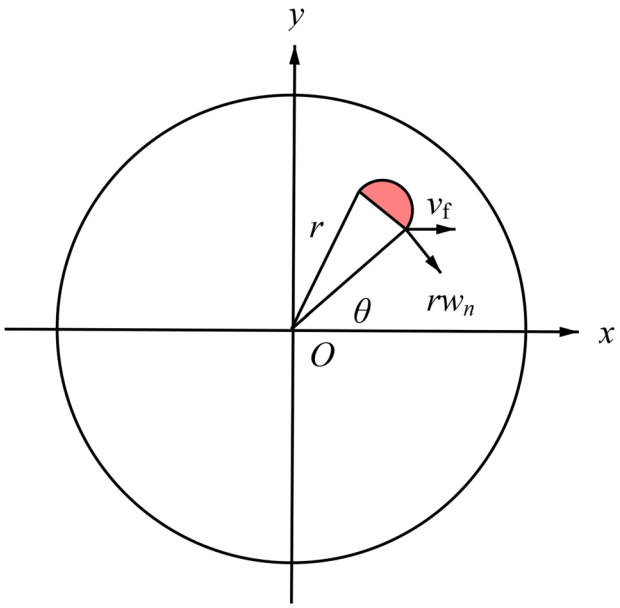


Fig. 11 Analysis of friction torque on polishing disc

$$M = \int_0^{R_p} \mathrm{d}M = \int_0^{R_p} R\mathrm{d}f = \int_0^{R_p}\int_0^{2} -\mu\frac{F_n}{\pi R_p}\frac{(v_f + r\omega_n\, sin\,\theta)\boldsymbol{i} - r\omega_n\, cos\,\theta\,\boldsymbol{j}}{\sqrt{(v_f + r\omega_n\, sin\,\theta)^2 + r^2{\omega_n}^2\, cos^2\theta}}\,\mathrm{d}r\mathrm{d}\,\theta \tag{24}$$

Substitute Eqs. (20) and (24) into Eq. (19) to get the E.

3.2 Parameter optimization based on genetic algorithm

Genetic algorithm is a heuristic search algorithm used for optimization. Its basic principle imitates the process of natural evolution and is widely used in parameter optimization in end milling [23], grinding [24], polishing [18], etc. Figure 12 shows the flow chart of using the genetic algorithm to optimize polishing parameters. The specific process is as follows:

1. Chromosomes are encoded by real number coding. In this paper, a chromosome containing two genes of polishing force F_n and feeding rate v_f is constructed. Discretize the specified polishing trajectory, and then optimize the parameters of each trajectory.
2. According to the actual experiment, the process constraints are designed, that is, the boundary of the parameters to be optimized, and then the appropriate population size is set. In this study, the constraints of F and v_f are as follows:

$$30\ \mathrm{N} \le F_n \le 50\ \mathrm{N} \tag{25}$$

$$1\ \mathrm{mm/s} \le v_f \le 4\ \mathrm{mm/s} \tag{26}$$

3. Input the generated polishing parameters into the material removal model with prior knowledge proposed above to predict the material removal depth, compare it with the expected material removal depth, and then calculate the fitness function G. The purpose of this paper is to ensure that the material removal depth is within the range of precision requirements and the processing energy consumption is minimal, so the fitness function shown in Eq. (27) is used.

$$G = \alpha_1 \cdot \int_0^t f\mathrm{d}t + E \tag{27}$$

$$f = \begin{cases} 1 & |h_p - h_0| > 0.05 \\ 0 & |h_p - h_0| \le 0.05 \end{cases} \tag{28}$$

where h_p represents the material removal depth predicted by the model, and h_0 represents the expected removal depth. Set a penalty item with a larger value to give priority to ensuring that the removal depth meets the requirements.

4. Generate a new generation of chromosomes with the best fitness through selection, crossover, and mutation. The selection operator adopts the roulette wheel selection. The arithmetic crossover method as shown in Eq. (29) is adopted.

$$\begin{aligned} C^1 &= a \times p_i^1 + (1-a)p_i^2 \\ C^2 &= a \times p_i^2 + (1-a)p_i^1 \end{aligned} \tag{29}$$

where C^1, C^2 represent the two newly generated children chromosomes, p_i^1, p_i^2 represent the parent chromosomes to be crossed, and a is a crossover constant.

Using the uniform variation shown in Eq. (30).

$$C_i^{'} = U_{\min} + u \times (U_{\max} - U_{\min}) \tag{30}$$

where C_i' is the gene to be mutated, $U_{\min}$ and $U_{\max}$ are the lower and upper bounds of the mutated gene, and u is a uniform random number between [0,1].

5. Repeat step (4) until the generations reach the set value and output the optimized parameters.

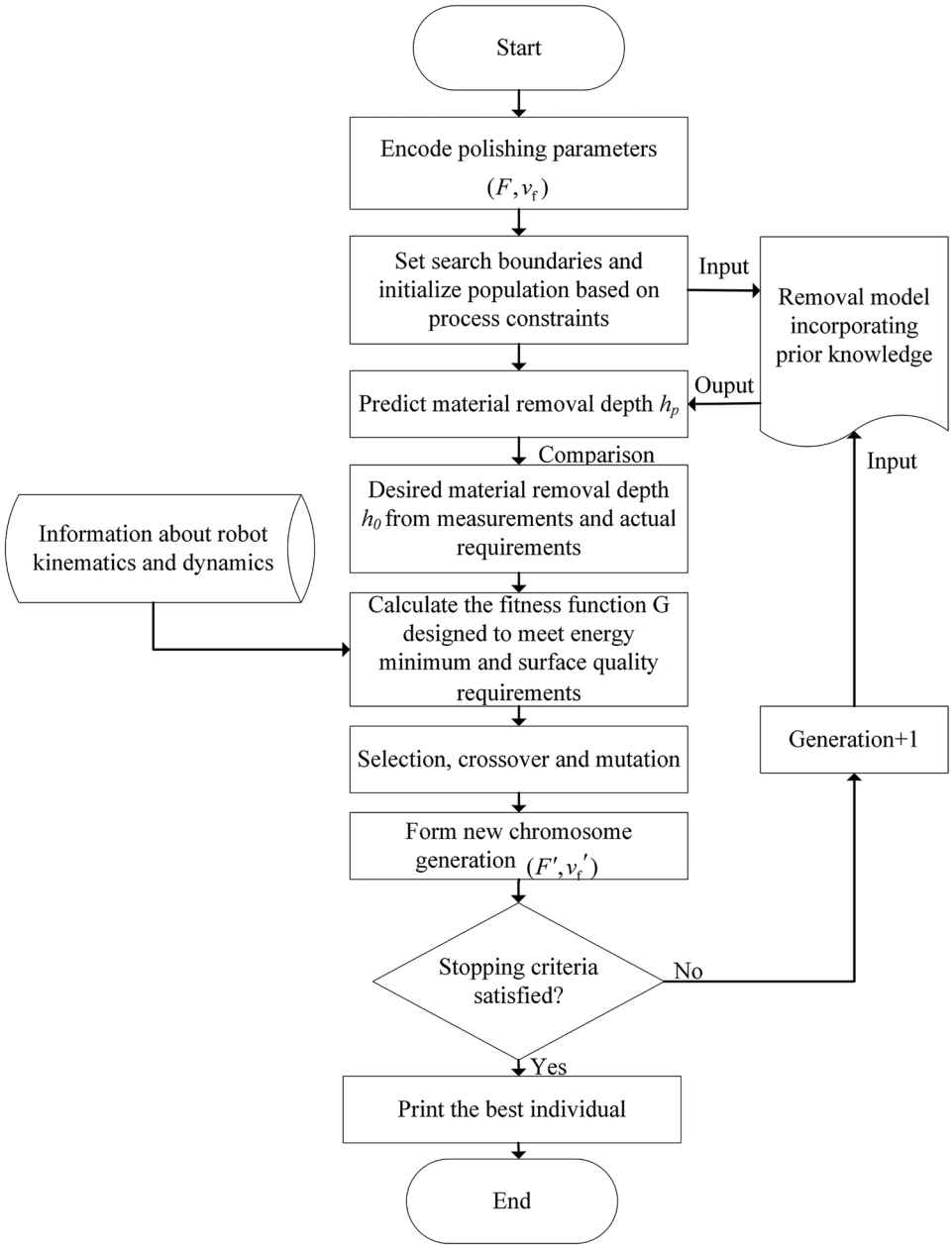
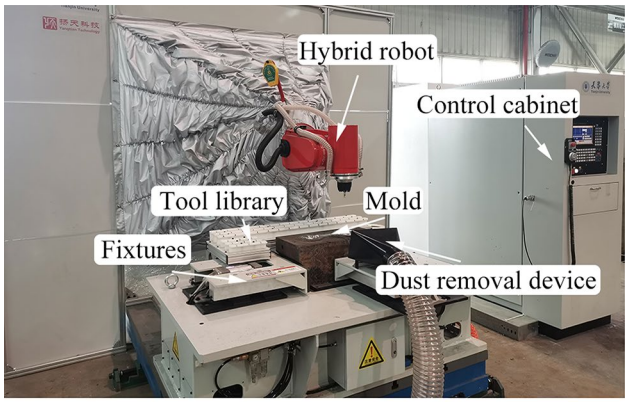


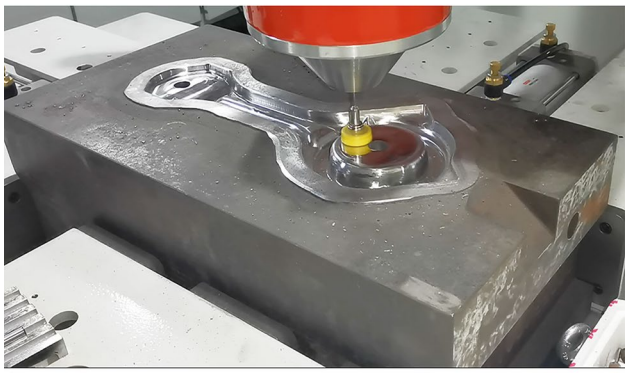
Fig. 12 Flow chart of genetic algorithm

4 Experimental results and discussion

The hybrid robot has been widely used in some processing fields such as spray-painting [25] and friction stir welding [26] due to their excellent characteristics of high stiffness and high precision [27], which is also an important reason why we use it for mold polishing. As shown in Fig. 13a, the experiment was carried out by using the hybrid robot polishing test bench. The test bench is composed of hybrid robot, control cabinet, tool library, fixtures, dust removal device, and mold. The related kinematics and dynamic parameters about the hybrid robot can be obtained from [28-30]. This hybrid robot has been used for polishing metal molds, and the algorithm proposed in this paper has been applied as shown in Fig. 13b and the video (Online Resource 2). Since the mold is too large to measure its surface topography on the 3D profiler, in order to verify the effectiveness of the proposed method, experiments are still carried out on specimens and the following comparative experiments are designed. The measured initial roughness parameter Ra (arithmetical mean roughness value) of the specimen is 2.4 μm, and the expected roughness Ra is 0.8 μm. The average material removal depth can be estimated to be 3.2 μm $(2.4 \times 2 - 0.8 \times 2)$ based on the initial surface profile and the desired surface profile [31], which is only a rough estimate to meet the processing effect. To prevent surface damage



(a) The test bench



(b) Processing display

Fig. 13 The hybrid robot polishing test bench and processing display

from over-polishing, we can set the average material removal depth to 3 μm. The experiment takes the AD segment in Fig. 3 as the optimized path. Through the optimization algorithm shown in "Sect. 3," the optimized polishing force and feed rate as shown in Fig. 14 can be obtained. The crossover constant a in Eq. (29) and mutation probabilities in the

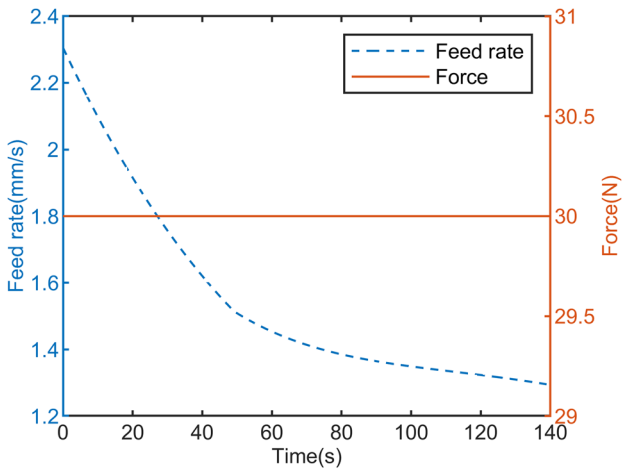


Fig. 14 The optimized polishing parameters

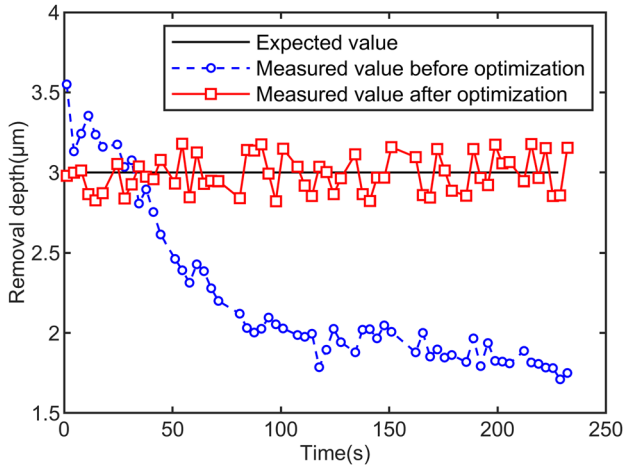


Fig. 15 Comparison of removal depth before and after parameter optimization

algorithm were set to 0.6 and 0.1, respectively, the maximum population generation was set to 100, and the population size was set to 50.

The surface of the specimen was also inspected with a 3D optical surface profiler to obtain the depth of removal during polishing. If the theoretical model described in [20] is used, when the removal depth is set to 3 μm and the given normal force is 40 N, the feed rate during polishing can be calculated to be 2.4 mm/s, which is a certain value. Using this set of polishing parameters for comparison, Fig. 15 shows the comparison of the material removal depth before and after optimization and the expected removal depth. It can be seen from Fig. 15 that the removal depth measured by the experimental group before the optimization is slightly larger than 3 μm in the early stage of polishing, and as the polishing tool wears, the removal depth gradually decreases and is always less than 3 μm, resulting in uneven removal and poor surface quality. Conversely, the removal

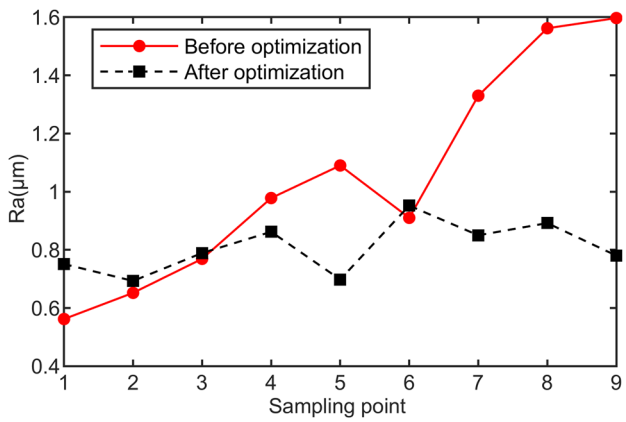


Fig. 16 Comparison of surface roughness before and after parameter optimization

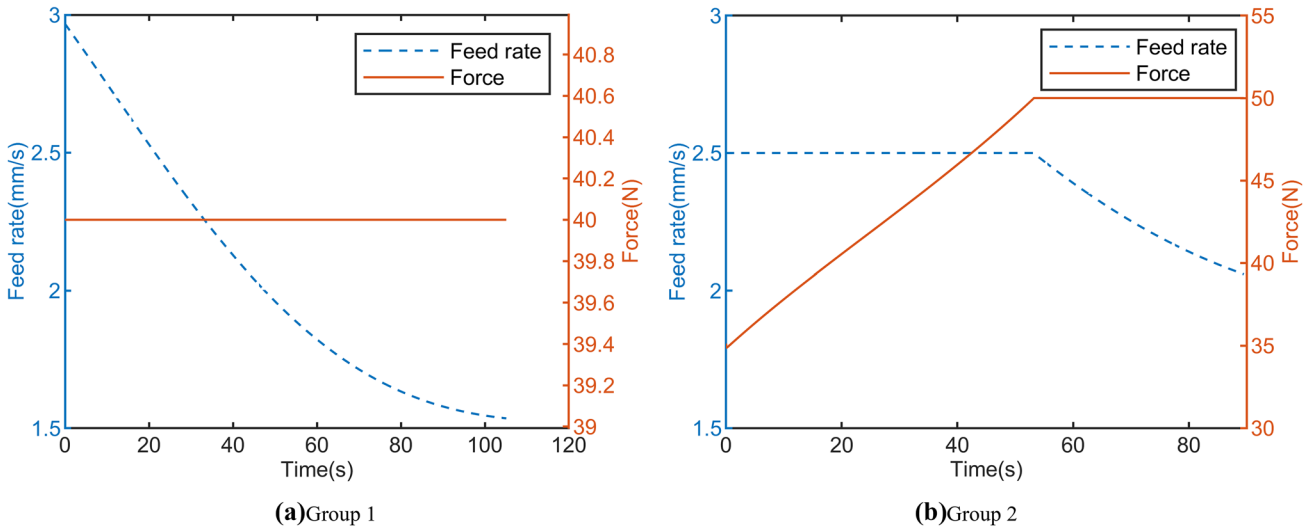


Fig. 17 Experimental groups for comparison

depth of the experimental group after optimization is more consistent with the expected value in the whole process, which can better compensate for the wear of sandpaper and ensure the uniformity of material removal and surface processing quality.

In addition, the Beijing Times TR200 high-precision surface roughness meter was used to measure the surface roughness of the specimen AD segment at 10 mm, 20 mm, 30 mm, 100 mm, 110 mm, 120 mm, 190 mm, 200 mm, and 210 mm from the starting point. Figure 16 shows the surface roughness of 9 sampling points before and after optimization. It can be seen from the figure that the surface roughness of the experimental group after optimization was relatively consistent in the whole process, the overall polishing effect was better, and more surfaces that meet the surface roughness requirements can be processed. On the contrary, the polishing quality of the unoptimized group decreased rapidly due to tool wear, and was inferior to the optimized experimental group for most of the time.

It is worth mentioning that the metal mold polishing aimed at in this paper only needs to meet the processing requirements, so a uniform material removal depth can be set according to the surface roughness. Note that some optical glass polishing with higher precision requirements faces the problem of uneven removal, and the method proposed in this paper is still applicable, as long as the actual required removal depth is substituted into the proposed model to solve.

Figure 17 shows the changes of the two groups of polishing parameters with time ensuring the removal depth is still 3 μm. The energy consumption of the two experimental groups and the optimized group can be calculated, respectively. It can be obtained by calculation that the energy consumption of the three experimental groups was 0.2467 kWh, 0.2550 kWh, and 0.2534 kWh, respectively. Compared to the other two groups, the optimized experimental group had a reduction of 3.34% and 2.7%, respectively. The actual polishing workload is obviously much larger than the AD segment path taken in the experiment, and the energy saved will be very considerable when the workload is accumulated. In summary, the experimental results show that the proposed polishing parameter optimization method can effectively ensure the polishing quality and reduce energy consumption.

5 Conclusion

In this paper, a method for optimizing robot polishing parameters considering time-varying wear is proposed. Firstly, the relationship between the removal depth and polishing parameters in the theoretical model is taken as prior knowledge, and combined with the GPR model that can learn the characteristics of tool wear and the influence of polishing parameters on k_p, a general material removal regression model incorporating prior knowledge is proposed. It greatly reduces the data dimension and data quantity required by the regression model, which is convenient for practical engineering applications. Compared with the theoretical model, SVR model, and GPR model proposed in other literature, it is proved that the GPR model incorporating prior knowledge has good prediction performance and significant advantages in small sample prediction. Based on the proposed model, the robot polishing parameters are optimized by a genetic algorithm. The experiments show that the proposed optimization method considering time-varying

wear can ensure polishing quality as well as reduction of energy consumption. Furthermore, because the tool wear is compensated, more surfaces that meet the requirements can be machined as much as possible before the tool fails. Since the currently used sandpaper still has a short service life after compensation, the future research interest is to study the wear variation of polishing tools with different materials to develop better mold polishing tools.

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Author contribution QZ came up with the original idea, designed the research, collected the data, and wrote the manuscript. The improvement of the article was completed under the guidance of JX. WC provides support for relevant experimental materials, techniques, and venues. HL made significant contributions to the analysis process. TH provided theoretical assistance. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets used or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability Some or all models, or code generated or used during the study, are available from the corresponding author by request.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no competing interests.

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